Malaria

Insects can help or harm humans. Bees pollinate plants used for food but many other insects can harm human health. When acting as disease-causing *vectors*, insects are harmful to humans and other animals. The female anopheles mosquito feeds on blood (unlike the male counterpart that feeds on nectar), and is probably the insect that has caused the most death and suffering in human history. Female mosquitoes need blood to produce eggs. In taking blood, the mosquito can pass on malaria to a person. This is called the *mechanical transfer* of an infective agent. In this case, females inject sporozoites, a motile infective form of parasitic protozoa that reproduce asexually in the new host.

1 What is *nectar*?

2 What is a *vector*?

Malaria is one of the world's most deadly diseases and kills millions of people every year. The mosquito does not directly cause malaria, but it infects humans with a protozoan parasite – *plasmodium* – that it carries within its own body. There are four species of malaria-causing plasmodium and one of them will be present in an infected person. The plasmodium finds its way into the mosquito when it bites someone infected with malaria. The parasite flows into the mosquito along with the blood that is drawn up from the bite. It will then reach a certain phase of development within the mosquito, after which it migrates to the salivary gland. There it mixes with saliva and anticoagulant used to 'thin' the blood of the host organism when it is bitten.

A female anopheles mosquito pierces the skin with a sharp mouthpart and injects saliva into the wound. Blood is sucked up by the mosquito's proboscis (tube). If the mosquito is infected with the protozoan parasite, then bites another human, the parasite will be injected into the person's bloodstream. Once there, the parasite migrates to the liver where it matures, replicates and causes a full-blown attack by infecting the red blood cells. Mosquitoes are *ectoparasites* – they live on the exterior of the host.

3 Define *parasite*.

4 Define *motile*.

5 Explain why a mosquito might benefit from injecting an anti-clotting agent into a person.

Protozoa are unicellular organisms. *Plasmodia* replicate in the liver for 2–4 weeks and then invade red blood cells, which eventually burst and release even larger numbers of plasmodia. Lysis, coupled with substances produced by the immune system, lead to the onset of fever and chills. The disease can be broken down to four stages. When the red blood cells have been infected to the point where there are so many spores within them that they burst, a malaria 'attack' occurs. Once infected with malaria, recurring attacks are not uncommon, particularly during times of stress. Symptoms will depend on the species of malaria involved but they include a combination of high fever and chills.

Identify the four stages of malaria below (6–9) using the terms *fever, incubation, shivering, sweating*.

6 Protozoa multiply but no symptoms are shown for the first one to two weeks.

7 Sufferers have an elevated body temperature, but feel cold for a short period.

8 Lasts for several hours, body temperature can rise above 41°C and red blood cells are bursting.

9 After the fever, the sufferer can sweat for several hours and usually recovers (about 1% of cases are fatal).

10 Without medical treatment the attacks return, even after person seems to have recovered. Name a drug used to treat malaria.

The name *malaria* comes from the Italian words *male*, meaning 'bad', and *aria* which means 'air'. Originally people thought malaria was caused by 'bad air'. Ronald Ross (1857–1932) was a British surgeon who studied malaria in India and came to the conclusion that malaria was carried by an insect vector – the female anopheles mosquito. Ross unravelled the complicated life cycle of the plasmodium parasite. This parasite bred in the stomach of the female mosquito, and then was carried by the bloodstream to the mosquito's salivary glands. When a victim was bitten the parasite was carried with the mosquito's saliva into the victim's bloodstream where it reached maturity. His work earned him the 1902 Nobel Prize for Medicine.

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11 Suggest why malaria is prominent in tropical and subtropical areas as opposed to other parts of the globe.

Mechanical transfer is not the only way insects can infect other animals. *Biological transfer* is a less random method. It involves a three-step process and cannot succeed if any step is missing or interfered with. The availability of the correct insect vector is the first step. The pathogen (disease-causing agent) must first replicate to sufficient numbers and reach a certain stage of development within the vector. The vector does not, however, have all of the necessary features for full infection to take place. Therefore, once sufficient numbers of the pathogen are prepared, it requires a host organism before full-blown infection is complete. The insect carries the pathogen to a host organism that provides what is required. The host organism is the 'final stop' and the one where maturation is complete. Different species of mosquito carry different forms of plasmodium and the combinations determine which animals will be affected by malaria. An estimated 125 million new cases of malaria are reported every year.

12 Why do female anopheles mosquitoes bite humans?

13 What is a *proboscis*?

14 In addition to transmitting protozoa, insects can infect animals with viruses and nematodes. What is a *nematode*?

15 Describe the difference between *mechanical transfer* and *biological transfer* of infective agents.

16 Pathogens spread by biological transfer usually require specific vectors. With this in mind, what would be the best way to reduce the rate of infection by these pathogens.

Cure and prevention

There have been many ways used to cure the symptoms of malaria or to prevent the disease. Quinine was originally extracted from the bark of a tree, but is now synthesised, and can relieve the symptoms of sweating and fever. Insect repellent is useful in ensuring that insects don't bite and hence can't infect. Netting or mesh over windows and beds can stop mosquitoes from reaching a human to bite them. Removing areas of stagnant water can stop insects breeding and insecticides can kill mosquitoes.

17 Put the information on curing and preventing malaria into the table below.

Method	Advantage	Disadvantage

Research

18 Malaria is not the only important disease transmitted by flying insects. Other examples include: sleeping sickness (trypanosomiasis), dengue fever, yellow fever and river-blindness (onchocerciasis). Investigate the causes and symptoms of one of the examples given above. Write a brief summary describing your choice.

19 People with *sickle cell anaemia* have a greater resistance to malaria than those who do not carry the sickle cell gene. Give brief details for sickle cell anaemia.

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